

How Water Travels in Plants!



GRADE 2 SCIENCE • THE MAGIC CELERY EXPERIMENT

Have you ever wondered how a tiny plant drinks water from the ground and sends it all the way up to its highest green leaves? Let's read a short story to find out!

Story Time: Why was the Tulsi Plant Sad?

Arjun forgot to water his home **Tulsi plant** for two whole days. The plant looked droopy, tired, and very sad! His Nani came outside and said, "*The plant is thirsty, beta!*"

Arjun quickly ran and poured a glass of water onto the dry soil. By evening, a miracle happened! The Tulsi plant was standing up tall, proud, and fresh again. Its leaves looked bright, happy, and green! 

The Big Science Question:

How did the water travel from the deep dark soil all the way up to the very top leaves? Let's become science detectives and solve this mystery together!

Meet the Plant Team (Key Words)

Plants have special parts that work like a team to move water:

1. Roots (जड़ / Jad)

Where it starts!

They live under the ground and soak up water from the soil. They work just like a bundle of **drinking straws** buried in the dirt!

2. Stem (तनाव / Tana)

The Plant Highway!

The long middle part of the plant. Inside, it has hundreds of tiny invisible tubes that carry water upward to the leaves.

3. Leaves (पत्ते / Patte)

The Final Stop!

The flat green parts at the top. They receive the water to stay fresh, crisp, and healthy under the warm sun.

4. Absorb

Science Action Word

To soak up or drink in water completely, just like how a kitchen sponge soaks up a spill!



The Magical Celery Experiment

LET'S MAKE THE INVISIBLE WATER JOURNEY VISIBLE!

Since we cannot see inside a normal plant stem, we are going to use a special vegetable called **celery** and colorful water to watch the water move with our own eyes!



Your Science Toolkit (Materials)

- **Fresh Celery Stalk:** Make sure it has plenty of pretty green leaves at the top!
- **Clear Glass or Jar:** It must be see-through so you can watch everything.
- **Food Coloring:** 10-15 drops of Red or Blue food color (Red shows up best!).
- **Water:** Plain water from the tap.
- **Scissors:** (For your Teacher or Parent's hands only!)



Easy Step-by-Step Guide

Step 1: Color the Water

Fill your clear glass 3/4 full with water. Add 10-15 drops of your food coloring and stir it well until it looks dark and colorful.

Step 2: Trim the Stalk (Adult Help!)

Ask your teacher or parent to carefully cut the bottom of the celery stalk at a slight angle. This makes it super easy for the celery to drink!

Step 3: Place and Stand Up

Put your celery stalk cut-side down into the colorful water. Lean it safely against the glass so it stands tall.

Step 4: Watch the Magic Happen!

Leave your jar in a safe spot. Check it after 30 minutes, then after 2 hours, and right before bed. What do you see changing?

What Will You Discover?

After a few hours or the next morning, you will see something amazing:

- The edges and tips of the green celery leaves will start turning **bright red or blue!**
- If an adult cuts across the stem, you will see a circle of tiny colorful dots—those are the plant's **drinking straws** working hard!



My Science Observation Journal

LITTLE SCIENTIST NAME:



My Celery Sketches

BEFORE (Day 1):

Draw your clean celery stalk here!

AFTER (Next Day):

Draw your colorful leaves here!



Fill in the Secret Path

Help the water find its way through the plant journey using the words: **Stem, Soil, Leaves, Roots.**

-  Water in the [1. _____]
-  Absorbed by the [2. _____]
-  Travels up the [3. _____]
-  Reaches the fresh green [4. _____]!



Fun Quick Quiz

Q1. What do roots act like under the ground?

Answer: _____

Q2. Why do flower bazaar sellers keep roses in buckets of water?

Answer: _____



Fun Challenge at Home!

Look around your house or balcony tonight. Find 2 different plants (like a Money Plant, Coriander/ Dhaniya, or Marigold flower). Remember that inside their stems, tiny water highways are working right now, even when you aren't looking! ✨